Solution:

1. Let d_t be the difference between a_t and b_t, then we can look at D as a third stream which has the values $d_t \in \{-1, 0, 1\}$ at time/step t. This of course follows from A and B where if $a_t = b_t$ then $d_t = 0$, and if $a_t \neq b_t$ then $d_t = 1$ if $a_t > b_t$ and $d_t = -1$ if $a_t < b_t$.
   Then H is by definition:
   $$H = \sum_{t=1}^{n} d_t^2$$
   This is the second frequency moment F2 of stream D, which means we can use AMS sketch to estimate H.

   The algorithm:
   We will use k independent random sign hash functions $h_i, i \in \{1, 2, \dots, k\}$.
   We will store k counters $Z_i, i \in \{1, 2, \dots, k\}$ all initialized to 0.
   Update at time t: for incoming elements at time t, calculate $d_t = a_t - b_t$, update every counter Z_i in the following manner $Z_i = Z_i + h_i(t) * d_t$.
   At the end of the stream return estimate of H:
   $$\widehat{H} = \frac{1}{k}\sum_{i=1}^{k} Z_i^2$$
   Correctness follows from AMS sketch.

2. For AMS we have seen at class that $Var(Z^2) \leq 2F_2^2 = 2H^2$, thus:
   $$Var\left(\frac{1}{k}\sum_{i=1}^{k} Z_i^2\right) = \frac{1}{k^2}\sum_{i=1}^{k} Var(Z_i^2) \leq \frac{2H^2}{k}$$
   Using Chebyshev's inequality:
   $$\Pr\left(\left|\frac{1}{k}\sum_{i=1}^{k} Z_i^2 - H\right| \geq \epsilon H\right) \leq \frac{Var\left(\frac{1}{k}\sum_{i=1}^{k} Z_i^2\right)}{\epsilon^2 H^2} = \frac{\frac{2H^2}{k}}{\epsilon^2 H^2} = \frac{2}{k\epsilon^2}$$
   If we set $k = \frac{20}{\epsilon^2}$, we get:
   $$\Pr\left(\left|\frac{1}{k}\sum_{i=1}^{k} Z_i^2 - H\right| \geq \epsilon H\right) \leq \frac{2}{k\epsilon^2} = 0.1$$
   From this and complementary probability it follows that:
   $$\Pr\left(\left|\frac{1}{k}\sum_{i=1}^{k} Z_i^2 - H\right| \leq \epsilon H\right) > 0.9$$

Each Z_i counts n elements so its space cost is $O(\log_2 n)$ bits.

We have $k = \frac{20}{\epsilon^2}$ counters Z_i, which $O(\epsilon^{-2})$ space.

Every hash function costs $O(\log_2 n)$ bits.

So overall we have a space cost of $O(\epsilon^{-2} \cdot \log_2 n)$ bits.

Thus, we are done. We have shown what's requested.

Lecture proof $Var(Z^2) \leq 2F_2^2$:

What is the variance of Z^2 ?

$$
\begin{aligned}
Var(r^2) &= E(r^4) - F_2^2 \\
E(r^4) &= E\left[(\sum_i x_i m_i)^2 (\sum_i x_i m_i)^2\right] \\
&= E[((\sum_i x_i^2 m_i^2) + (2\sum_{i<j} x_i x_j m_i m_j))^2] \\
&= E[(\sum_i x_i^2 m_i^2)^2] + 4E[(\sum_{i<j} x_i x_j m_i m_j)^2] + 4E[(\sum_i x_i^2 m_i^2)(\sum_{i<j} x_i x_j m_i m_j)]
\end{aligned}
$$

The last term is 0 since every pair of variables x_i and x_j are independent. Since $x_i^2 = 1$, the first term is F_2^2.

$$
\begin{aligned}
Var(r^2) &= E(r^4) - F_2^2 = 4E[(\sum_{i<j} x_i x_j m_i m_j)^2] \\
&= 4E[\sum_{i<j} x_i^2 x_j^2 m_i^2 m_j^2] + 4E[\sum_{i<j<k<l} x_i x_j x_k x_l m_i m_j m_k m_l]
\end{aligned}
$$

Again, the last term is 0 since every set of 4 random variables is independent of each other. Therefore,

$$
Var(r^2) = 4\sum_{i<j} m_i^2 m_j^2 \leq 2F_2^2
$$